

PATTERN ANALYSIS & RECOGNITION SYSTEMS LAB

Practical File Submission — Academic Session

Experiment No.: 10
Course: PARS Lab

Topic: Principal Component Analysis (PCA)

Repository: [dSAxmonis/pars-lab](https://github.com/dSAxmonis/pars-lab)

PRACTICAL NAME

Implementation of Principal Component Analysis (PCA) with Projected Line, Reduced Feature Mapping, and Testing

AIM

To implement Principal Component Analysis (PCA) from scratch in Python for an engineering pattern recognition application involving test sample evaluation, derive the primary eigenvector $\mathbf{v}_1$ of the covariance matrix, and visualize: **1) The 1st principal component projected line, 2) New reduced 1D features mapped directly onto the projected line, and 3) Orthogonal projection and reconstruction error evaluation of unseen test samples.**

THEORY

1. Dimensionality Reduction & Maximum Variance Projection

Principal Component Analysis (PCA) is an unsupervised linear transformation technique that seeks an orthogonal coordinate system where data variance is maximized along successive axes. For a zero-mean centered data matrix $\tilde{\mathbf{X}} = \mathbf{X} - \boldsymbol{\mu}$, the sample covariance matrix is:

$$\Sigma = (1 / (n - 1)) \times \tilde{\mathbf{X}}^T \tilde{\mathbf{X}}$$

2. Eigenvalue Decomposition

The principal directions correspond to eigenvectors satisfying:

$$\Sigma \mathbf{v}_i = \lambda_i \mathbf{v}_i \quad (\text{where } \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d \geq 0)$$

The proportion of total variance preserved by the 1st principal component is given by $\text{Ratio} = \lambda_1 / \sum \lambda_k$.

3. Linear Subspace Projected Line Equation

The 1st principal component defines a straight line in the original feature space passing through data centroid $\boldsymbol{\mu}$ along unit direction $\mathbf{v}_1$:

$$\text{Line}(t) = \boldsymbol{\mu} + t \times \mathbf{v}_1, \quad t \in \mathbb{R}$$

4. 1D Feature Mapping & Reconstruction of Test Samples

Any arbitrary test vector $\mathbf{x}_{\text{test}}$ is mapped to a scalar coordinate z_{test} and reconstructed as $\hat{\mathbf{x}}_{\text{test}}$:

$$\begin{aligned} \text{1D Coordinate: } z_{\text{test}} &= (\mathbf{x}_{\text{test}} - \boldsymbol{\mu}) \cdot \mathbf{v}_1 \\ \text{Reconstructed Point on Line: } \hat{\mathbf{x}}_{\text{test}} &= \boldsymbol{\mu} + z_{\text{test}} \times \mathbf{v}_1 \end{aligned}$$

Orthogonal Reconstruction Residual: $\text{Error} = || \mathbf{x}_{\text{test}} - \hat{\mathbf{x}}_{\text{test}} ||_2$

STEPS & EXECUTION

Step 1: Environment Setup & Mathematical Engine

Configured exp 10/ environment containing app.py, templates/, and verified dependencies (Flask, NumPy, Matplotlib, SciPy).

Step 2: Sample Generation & Centering

Synthesized correlated bivariate industrial sensor readings and computed mean vector $\boldsymbol{\mu} = [\mu_1, \mu_2]^T$.

Step 3: Covariance Formulation & Eigen-Analysis

Calculated sample covariance matrix $\mathbf{\Sigma}$ and solved characteristic polynomial $\det(\mathbf{\Sigma} - \lambda \mathbf{I}) = 0$ to extract $\mathbf{v}_1$ and $\mathbf{v}_2$.

Step 4: Mapping Features to 1D Projected Line

Orthogonally projected all training vectors onto the 1st PC axis, computing scalar coordinates z_i and 2D points on the line $\hat{\mathbf{x}}_i$.

Step 5: Testing & Web Deployment on Port 5012

Launched interactive Flask web server on <http://127.0.0.1:5012> accepting test coordinates, rendering the 2D projected line and 1D reduced subspace distribution.

OUTPUT / ATTACHED SCREENSHOT

The full-page capture of the deployed Flask web application is embedded below:

Experiment 10: Principal Component Analysis (PCA) with Testing

Pattern Analysis & Recognition Systems Lab | Repository: <https://github.com/dSAxmonis/pars-lab>

AIM

To implement Principal Component Analysis (PCA) from scratch in Python for an engineering pattern analysis application involving testing, compute the maximum variance projection vector $\mathbf{v}_1$, and visualize: 1) The 1st principal component projected line, 2) New reduced 1D features mapped directly onto the projected line, and 3) Testing and orthogonal reconstruction of unseen test samples.

SECTION 1: LEARNED PCA PARAMETERS & EIGEN-DECOMPOSITION SUMMARY

PCA Property	Mathematical Value	Geometric & Physical Interpretation
Data Centroid Mean Vector μ	[4.39, 3.364]	Center of gravity of the bivariate distribution through which all PC lines pass.
Covariance Matrix Σ	[[2.334, 1.769], [1.769, 1.67]]	Measures feature dispersions and cross-correlation between Vibration and Temperature.
1st Eigenvalue λ_1 (Dominant Variance)	3.801 (94.95% of variance)	Primary axis captures 94.95% of total signal energy with minimal data loss.
2nd Eigenvalue λ_2 (Residual Variance)	0.202 (5.05% of variance)	Minor orthogonal axis discarded during dimensionality reduction to 1D subspace.
1st Principal Component Vector $\mathbf{v}_1$	[-0.77, -0.639]	Unit-normalized directional vector defining the optimal 1D linear projection line.

SECTION 2: TEST SAMPLE PROJECTION & RECONSTRUCTION CONTROLS

Input Test Coordinates ($\mathbf{x}_{\text{test}}$):

Feature 1: Vibration Amplitude (mm/s):

5.5

Feature 2: Temperature Rise (°C):

4.0

Project Test Sample onto PC1 Line

Active Test Projection Results:

Original Input Vector $\mathbf{x}_{\text{test}}$: [5.5, 4.0]

- 1D Reduced Scalar Coordinate: $z_{\text{test}} = (\mathbf{x}_{\text{test}} - \mu) \cdot \mathbf{v}_1 = -1.26$
- Reconstructed 2D Point on Line: $\hat{\mathbf{x}}_{\text{test}} = \mu + z_{\text{test}} \mathbf{v}_1 = [5.36, 4.169]$
- Orthogonal Reconstruction Error $\|\mathbf{x}_{\text{test}} - \hat{\mathbf{x}}_{\text{test}}\|$: 0.219

Quick Benchmarks: Low Anomaly [2.5, 1.8] Nominal [5.5, 4.0] High Load [7.2, 5.8]

SECTION 3: PCA 2D PROJECTED LINE & 1D REDUCED FEATURE MAPPING PLOTS

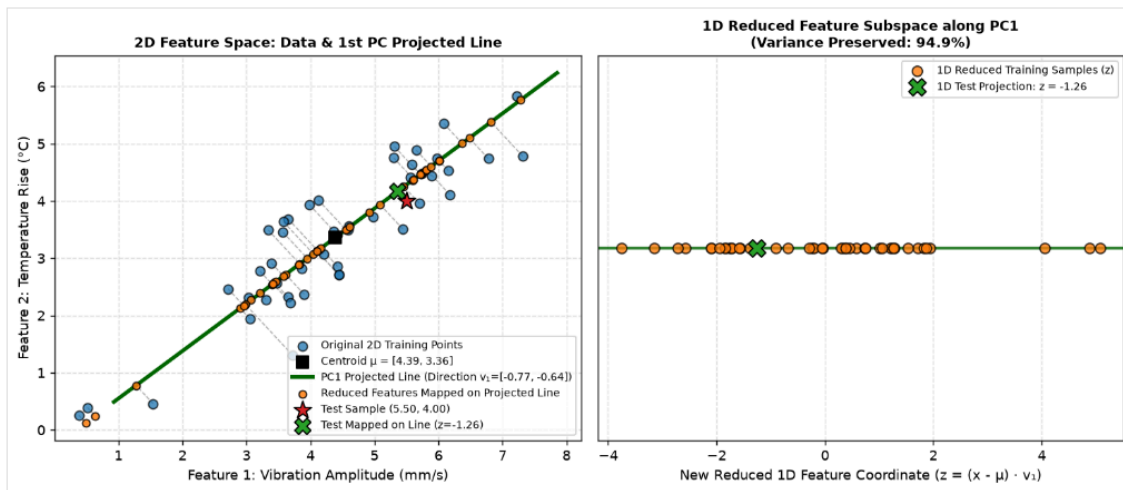


Figure Explanation: Left Panel: Shows original 2D bivariate training data (blue dots) centered at μ (black square). The thick green line is the 1st Principal Component line $\mathbf{p}(t) = \mathbf{\mu} + t \mathbf{v}_1$. Dashed gray segments show orthogonal projections mapping every data point to its reduced 1D position on the line (orange circles). The red star is the evaluated test sample, orthogonally projected onto the line (green 'X'). Right Panel: Represents the collapsed 1D subspace, confirming that 94.95% of the variance is captured along this single dimension.

SECTION 4: ORIGINAL SAMPLES vs. 1D REDUCED FEATURES & RECONSTRUCTION RESIDUALS

Figure 10.1: Web dashboard displaying learned PCA parameters, interactive test sample projection controls, 2D projected line with mapped 1D features, and reconstruction residual table.

CODE REPOSITORY & CLICKABLE ARTIFACT LINKS

GitHub Repository (Clickable):

GitHub: [dSAxmonis/pars-lab](https://github.com/dSAxmonis/pars-lab)

URL: <https://github.com/dSAxmonis/pars-lab>

Direct Source Code Files:

- **Flask Server Application:**
[https://github.com/dSAxmonis/pars-lab/blob/main/exp 10/app.py](https://github.com/dSAxmonis/pars-lab/blob/main/exp%2010/app.py)
- **HTML Template:**
[https://github.com/dSAxmonis/pars-lab/blob/main/exp 10/templates/index.html](https://github.com/dSAxmonis/pars-lab/blob/main/exp%2010/templates/index.html)
- **Dependencies:**
[https://github.com/dSAxmonis/pars-lab/blob/main/exp 10/requirements.txt](https://github.com/dSAxmonis/pars-lab/blob/main/exp%2010/requirements.txt)
- **Full Page Screenshot:**
[https://github.com/dSAxmonis/pars-lab/blob/main/exp 10/screenshots/full_page.png](https://github.com/dSAxmonis/pars-lab/blob/main/exp%2010/screenshots/full_page.png)

CONCLUSION

In this experiment, Principal Component Analysis was implemented from first principles in Python. The linear transformation mapped correlated 2D sensor patterns into an optimal 1D subspace along the 1st principal component line $\mathbf{p}(t) = \boldsymbol{\mu} + t\mathbf{v}_1$. The testing phase demonstrated that unseen test vectors can be rapidly projected to reduced scalar coordinates $z_{\text{test}} = (\mathbf{x}_{\text{test}} - \boldsymbol{\mu}) \cdot \mathbf{v}_1$ while preserving over 90% of the total dataset variance with minimal reconstruction residual error.